ArchCheck · 英文汇报讲稿（2026-09-01）

> 配套英文 PPT： `本周进展汇报_2026-09-01_EN.pptx` （8 页，白底 · 宋体 · 四号）
> 每页给一段可照读的英文口语讲稿 + 一行中文提示（万一要切中文，或帮你回忆本页要点）。
> 全程约 6-7 分钟。

Slide 1 · 标题

English

> Good afternoon. This is our weekly progress report for **ArchCheck**, an AI tool that checks a building model against fire-safety and building codes. Today I will cover three lines of work: reading and processing the model, mapping rules to the code, and a survey of B-Rep geometry. After that, I will show you a live demo.

中文提示：今天讲三件事：模型读取与信息处理、规则对应文献、B-Rep 调研，最后现场演示。

Slide 2 · 回应上周老师的问题

English

> Last week you asked us three things. First — does the IFC file carry B-Rep? We checked. Yes, the main sample models have B-Rep built in. So we can do exact geometry without building a CSG kernel. Second — can the knowledge base really judge a building, not just search articles? Now yes. After the extractor upgrade, the door–room link went from 6 rooms to 98 rooms out of 100. Third — what is our search style? We use keywords first, and semantic search as a fallback.

中文提示：三个问题逐一答：①IFC 带不带 B-Rep？带，无需 CSG 内核；②知识库能不能真判定？能，门-房间关联 6→98/100；③检索口径？关键词优先、语义兜底。

Slide 3 · 主线一：模型读取与信息处理

English

> Let me start with the model side. First, the door–room link. The old problem: some rooms had no solid geometry, so we could not tell which room a door belonged to. Our fix: give every room a floor footprint from its bounding box, include the rotation in the placement matrix, and prefer real boundary data when it exists. The result: escape distance can now be judged for 98 rooms, and we found two real violations — two rooms that have no exit door at all.
>
> Second, fire-hydrant spacing. We taught the extractor to read fire-pipe geometry, take the centre point of each pipe, and compute the distance to its nearest neighbour. This moves us from reading attributes to reading geometry and computing distances. On the office sample, six fire pipes are 13 metres apart — well under the 30-metre limit.

中文提示：门-房间关联方法（bbox 足迹 + 放置矩阵旋转 + 边界优先）→ 98 房可判、查出 2 处真违规；消火栓间距方法（消防管质心最近邻）→ 从读属性升级到读几何算距离，office 6 根管 13m ≤ 30m。

Slide 4 · 主线二：规则扩量

English

> Why can we now match the code? Because our rules hold real thresholds taken from the standards: GB 55037, GB 50016, GB 50974, and the Singapore Fire Code. Let me explain the principle — three points. One: use real fields only. A rule only uses fields the extractor really outputs, so we have no dead rules. Two: scope by condition. A rule judges only matching elements — for example, only fire pipes. Three: cite the code. Every rule points to the exact clause, and the threshold is verified from the source text — a door must be 0.9 metres wide, hydrants 30 metres apart.
>
> The effect: a question finds a clause, the clause maps to a rule, and the rule makes a real judgement on the model. The knowledge base has moved from search to a basis for judgement, and the rule set grew from 21 to 47 rules per region.

中文提示：规则扩量的原理三要素：①真实字段（对齐提取器、无死规则）②条件限定范围（如只判 fire 管）③规范出处（code_ref 对应条款、阈值按原文核实 0.9m/30m）。作用：提问→条文→规则→真判定；21→47 条/地区。

Slide 5 · 主线三：B-Rep 调研

English

> We surveyed four sample models and counted the B-Rep solids. Three of them carry B-Rep by default — Schependomlaan has more than four thousand. This changed our plan. Originally we planned to use CSG plus Boolean operations. But CSG keeps the modelling history and needs a self-built solid kernel, which is expensive. Let me make the difference clear: CSG is a parametric construction that keeps history; B-Rep is an explicit shell — surfaces that close into a solid. Since B-Rep is already there, we use it directly. The divergence theorem, with signed tetrahedra, gives us the exact volume — we verified it on columns, beams and slabs. So Step 3 is a B-Rep / mesh mix with a precision note. We do not need a CSG kernel.

中文提示：调研计数（Schep 4202 / office 362 / FZK 140 / Duplex 0）→ B-Rep 主流自带；原本计划 CSG+布尔、麻烦在自研内核；CSG=参数化保留历史，B-Rep=显式外壳；散度定理（有符号四面体）算体积已验证 → 台阶③ 走 B-Rep/网格混合，不用 CSG。

Demo · 现场演示讲稿（跑 demo 用，约 2 分钟）

开场 · 先讲清楚这个 py 的来由

> Why did we write this small script? Our platform UI shows the final review report, but not the inside of a judgement — which clause was hit, which rule it matched, and why. So we made a small command-line demo (`scripts/demo_reasoning.py`) that prints the whole chain: **question → found clause → matched rule → judgement**. These are its outputs on two models.

中文提示：先说明——平台只给最终报告、看不到判断过程，所以做了命令行 demo 把链路打出来（提问→条文→规则→判定），下面是运行结果。

屏幕 A · 公寓（CN · Schependomlaan）—— 完整链路 + 诚实上报

> First, the apartment model. Door width found **11 real violations**: door 2.01 is 0.795 m, below the 0.9 m minimum. Now the honest part — the model has no electrical equipment, so emergency lighting reports **"not checked"**; and it has 60 pipes but no fire pipes, so hydrant spacing reports **"no condition match"**. It never fakes a pass.

屏幕旁边英文注解：

> Apartment — Full Chain, Honest Reporting
>
> - Door width: 11 real violations — door 2.01 is 0.795 m, below the 0.9 m minimum
> - No electrical equipment → emergency lighting reports "not checked"
> - 60 pipes but no fire pipes → hydrant spacing reports "no condition match"

中文提示：公寓屏——门宽 11 处违规（2.01 门 0.795m<0.9m）+ 两处"未检查"（无电气 / 有管无 fire 管），它不假装通过。

屏幕 B · 办公楼（CN · office_3f）—— 数据补齐，规则激活

> Same questions, a data-complete office model. Emergency lighting finds **3 real violations** — corridors without emergency lights. Fire hydrants: **6 pipes, spacing 13.0 m ≤ 30 m → all pass**. The point: better model reading wakes the rules up automatically.

屏幕旁边英文注解：

> Office — Same Questions, Data Complete
>
> - Emergency lighting: 3 real violations (corridor without emergency lights)
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中文提示：办公楼屏——应急照明 3 处真违规、消火栓 6 根 13m≤30m 全过 → 数据补齐规则就自动激活。

过渡句（放两屏中间或收尾）

> Better model reading → rules wake up automatically. No data = honest "not checked"; full data = real judgement.

中文提示：模型读取越好，规则越活；没数据就诚实"未检查"，有数据就真判定。

Slide 7 · 本周提取器升级的效果

English

> This table shows what this week's upgrade changed on one sample. Door width: last week the data was missing; now we found 11 violations. Escape distance: last week we could judge only 6 rooms; now all 100. Exit count: last week there were 2 false alarms; now it is clean, because we fixed the false positives. In short — we moved from "cannot judge" to "real judgement".

中文提示：三行对比（门宽 数据缺失→11 处违规；疏散距离 只能判 6 房→100 房；出口数量 误报 2 处→通过）→ 从判不了到判得动。

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（可选）屏幕 C · SG —— 双标准一句带过

> One rule set can switch standards — the same door needs 0.9 m in CN, 0.85 m in SG Fire Code. Rules are traceable and switchable.

中文提示：可选——同一道门 CN 0.9m ↔ SG 0.85m（FC23），规则可追溯、可换标准。

附：讲的时候注意

- 数字都能复算：98/100、11 处、13.0m、30m、0.9m、6 根、4202、21→47。
- 「未检查 / 条件无匹配」这类诚实标注不上 PPT，但讲到 demo 时可以口头带过一句：「这套规则如果模型没喂到数据，会如实标成 not checked，不会假装通过。」